Introduction

Network meta-analysis combines direct evidence from head-to-head trials with indirect evidence inferred through common comparators. The validity of this synthesis hinges on the consistency assumption: across closed loops in the network, the direct and indirect estimates of any pairwise contrast must agree up to sampling error. When they disagree systematically — when, for example, the direct A-vs-C estimate from head-to-head trials differs materially from the indirect estimate inferred via B — the network is inconsistent, and the pooled NMA estimates cannot be interpreted at face value. Detecting and characterising inconsistency is therefore a mandatory step in any rigorous NMA workflow, and modern guidance (NICE DSU, ISPOR, PRISMA-NMA) requires that consistency be reported alongside the league table and SUCRA results.

Prerequisites

A working understanding of network meta-analysis structure (treatments, designs, contrasts), the difference between direct and indirect evidence, and the transitivity assumption that underpins indirect comparisons.

Theory

Three complementary tests address different scales of inconsistency. The loop-specific Bucher test, for a triangle A-B-C, compares the direct A-vs-C estimate against the indirect estimate $\hat{\theta}_{AB} + \hat{\theta}_{BC}$ via a Wald-type difference. Node-splitting (Dias et al., 2010) generalises this for any contrast: it splits the evidence for one contrast into direct-only and indirect-only components and tests whether they agree. The design-by-treatment interaction model (Higgins et al., 2012) provides a global test via the generalised-inconsistency framework, comparing the consistent NMA model against an unconstrained model that allows different contrasts to take different values across study designs.

Assumptions

The network contains at least one closed loop (otherwise consistency cannot be tested), transitivity holds in principle, and the underlying meta-analytic model is correctly specified. Inconsistency tests have low power when loops are short or sparsely populated; absence of detected inconsistency does not prove consistency.

R Implementation

```r
library(netmeta)

set.seed(2026)
net_df <- data.frame(
  study = c("S1", "S1", "S2", "S2", "S3", "S3", "S4", "S4",
            "S5", "S5", "S6", "S6"),
  treat = c("A", "B", "A", "C", "B", "C", "A", "D",
            "B", "D", "C", "D"),
  TE    = c(0, -0.3, 0, -0.5, 0.4, 0.2, 0, -0.2,
            0.1, 0.3, 0, 0.4),
  seTE  = rep(0.25, 12)
)

p_df <- pairwise(treat = treat, TE = TE, seTE = seTE,
                 studlab = study, data = net_df)
net  <- netmeta(p_df, common = FALSE, reference = "A")

netsplit(net)

decomp.design(net)
```

Output & Results

`netsplit()` returns per-comparison direct vs indirect estimates, their difference, and a Wald test on the difference. `decomp.design()` decomposes the heterogeneity statistic Q into within-design and between-design components and reports the global design-by-treatment interaction test. Non-significant results in both diagnostics support the consistency assumption.

Interpretation

A reporting sentence: "Node-splitting (`netsplit`) showed agreement between direct and indirect estimates for all six pairwise comparisons (smallest two-sided p-value = 0.15). The global design-by-treatment interaction test (`decomp.design`) was non-significant ($p = 0.36$), and the between-design heterogeneity component was negligible ($Q_{\text{inc}} = 1.2$, df = 3). The consistency assumption is therefore supported, and the random-effects league table can be interpreted as reflecting the joint direct and indirect evidence." Always report both tests.

Practical Tips

- Always test consistency; reporting an NMA without any consistency check is no longer acceptable in major journals or HTA submissions.
- If inconsistency is detected, investigate study-level effect modifiers (population age, baseline severity, dose, year, risk of bias) that differ across designs; meta-regression within the NMA framework can then partially resolve the inconsistency.
- `netsplit` provides per-comparison tests (high resolution, low power per test); `decomp.design` provides the global test (low resolution, higher power overall) — they are complementary, not redundant.
- For simple triangles with sufficient direct evidence, the Bucher test is a quick first-pass diagnostic; it is what node-splitting reduces to in the three-treatment case.
- Inconsistency can be local (a single loop) or global (the whole network); the diagnostic patterns differ, and the discussion section should reflect which kind was found.
- Bayesian implementations in `gemtc` and `multinma` provide UME (unrelated mean effects) consistency models that compare against the consistent NMA via DIC; this is the Bayesian analogue of the design-by-treatment test.

R Packages Used

`netmeta::netsplit()` and `netmeta::decomp.design()` for the canonical frequentist consistency diagnostics, alongside `netmeta()` itself; `gemtc` and `multinma` for Bayesian NMA with built-in UME and node-splitting; `pcnetmeta` for component-NMA inconsistency analysis; `BUGSnet` for combined Bayesian NMA workflow including consistency reporting.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Systematic reviews and PRISMA
- Meta-analysis basics
- Network meta-analysis